

EXP:1 Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim:

To implement and demonstrate the FIND-S Algorithm for finding the most specific hypothesis that is consistent with the given training examples.

Algorithm Steps:

1. Initialize the hypothesis as the most specific hypothesis.
2. For each positive training example:
 - Compare it with the current hypothesis.
 - If an attribute differs, replace it with '?'.
3. Ignore all negative training examples.
4. Output the final hypothesis.

Program:

```
import pandas as pd

data = {
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],
    'AirTemp': ['Warm', 'Warm', 'Cold', 'Warm'],
    'Humidity': ['Normal', 'High', 'High', 'High'],
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],
    'Water': ['Warm', 'Warm', 'Warm', 'Cool'],
    'Forecast': ['Same', 'Same', 'Change', 'Same'],
    'EnjoySport': ['Yes', 'Yes', 'No', 'Yes']
}

df = pd.DataFrame(data)

print("Training Data:\n")

print(df)
```

```

concepts = df.iloc[:, :-1].values
target = df.iloc[:, -1].values
def find_s(concepts, target):
    hypothesis = None
    for i in range(len(target)):
        if target[i] == "Yes":
            if hypothesis is None:
                hypothesis = concepts[i].copy()
            else:
                for j in range(len(hypothesis)):
                    if hypothesis[j] != concepts[i][j]:
                        hypothesis[j] = '?'
    return hypothesis
final_hypothesis = find_s(concepts, target)
print("\nMost Specific Hypothesis:")
print(final_hypothesis)

```

Output:

```

... Training Data:
   Sky AirTemp Humidity  Wind Water Forecast EnjoySport
0  Sunny   Warm  Normal  Strong  Warm   Same      Yes
1  Sunny   Warm   High  Strong  Warm   Same      Yes
2  Rainy   Cold   High  Strong  Warm  Change     No
3  Sunny   Warm   High  Strong  Cool   Same      Yes

Most Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' 'Same']

```

Result:

The FIND-S algorithm was successfully implemented using Python.

